

Why the AVA and AVB Drive Onsets Are SNICs

Equilibrium stability and infinite-period scaling in the 17-cell segment

Distribution of Oscillators Project

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Abstract

We classify the onset of rhythmic activity in the 17-cell *C. elegans* segment as either AVA or AVB command current is increased from zero. In both branches, the resting equilibrium terminates when a real eigenvalue reaches zero, a finite-amplitude periodic orbit exists immediately above the same current, and its period follows the inverse-square-root divergence expected at a saddle-node bottleneck. The combined evidence identifies both onsets as saddle-node-on-invariant-circle (SNIC) bifurcations rather than Hopf or saddle-homoclinic transitions.

1 Question and model

The question is not merely where oscillation begins, but what local and global dynamics create it. We therefore combine equilibrium continuation, Jacobian eigenvalues, direct periodic-orbit measurements, and near-onset scaling.

The calculation uses the reduced 34-state (V, w) representation of the full 17-cell segment. Synaptic fatigue is excluded because it currently evolves without multiplying the synaptic signal and therefore cannot affect the voltage-recovery bifurcation. AVA and AVB are studied separately: one command current is varied while the other is held at zero.

The controlled parameter set is

$$\begin{aligned} G_{\text{syne}} &= 0.07 \text{ nS}, & G_{\text{syni}} &= 0.05 \text{ nS}, & G_{\text{gap}} &= 0.03 \text{ nS}, \\ k_{\text{syn}} &= 0.25 \text{ mV}^{-1}, & V_{\text{th}} &= -52 \text{ mV}, & \beta &= 1.03, & \tau_w &= 400 \text{ ms}. \end{aligned}$$

The value $\tau_w = 400 \text{ ms}$ is the canonical baseline selected on 2026-08-25 from a controlled AVA/AVB dorsoventral-phase sensitivity sweep. The original 2026-08-18 calculation used 200 ms; recomputation at 400 ms preserves the equilibrium fold locations and SNIC classification while changing the realized cycle timescale and muscle output.

2 Numerical result

The resting equilibrium terminates at

$$\boxed{I_{\text{SN}}^{\text{AVA}} = 1.68143 \text{ pA}}, \quad \boxed{I_{\text{SN}}^{\text{AVB}} = 1.59643 \text{ pA}}. \quad (1)$$

Table 1 summarizes the independent numerical signatures.

Table 1: Evidence used to classify the lower command-drive bifurcations.

Diagnostic	AVA	AVB
Fitted onset current I_{SN} (pA)	1.68143	1.59643
Fold fit R^2 : $\lambda^2 \propto I_{\text{SN}} - I$	0.99997	> 0.999999
Functional amplitude observable	Δ_{DV}	Δ_{DV}
Near-onset DV amplitude (mV)	≈ 47.6	≈ 47.7
Period at $\Delta I = 0.025$ pA (s)	3.37	3.46
Period at $\Delta I = 0.0001$ pA (s)	18.08	18.66
Fit R^2 : $T = A/\sqrt{\Delta I} + B$	0.99950	0.98774

3 Evidence 1: the equilibrium has a saddle-node

Figure 1 follows the stable resting state as command drive increases. The branch terminates at a fold. In the lower panels, the critical eigenvalue is purely real and approaches zero. The near-fold relation

$$\lambda^2 \propto I_{\text{SN}} - I \quad (2)$$

has essentially unit coefficient of determination for both commands, as expected for a generic saddle-node. This establishes the “SN” part of SNIC.

AVA and AVB command onset: saddle-node evidence

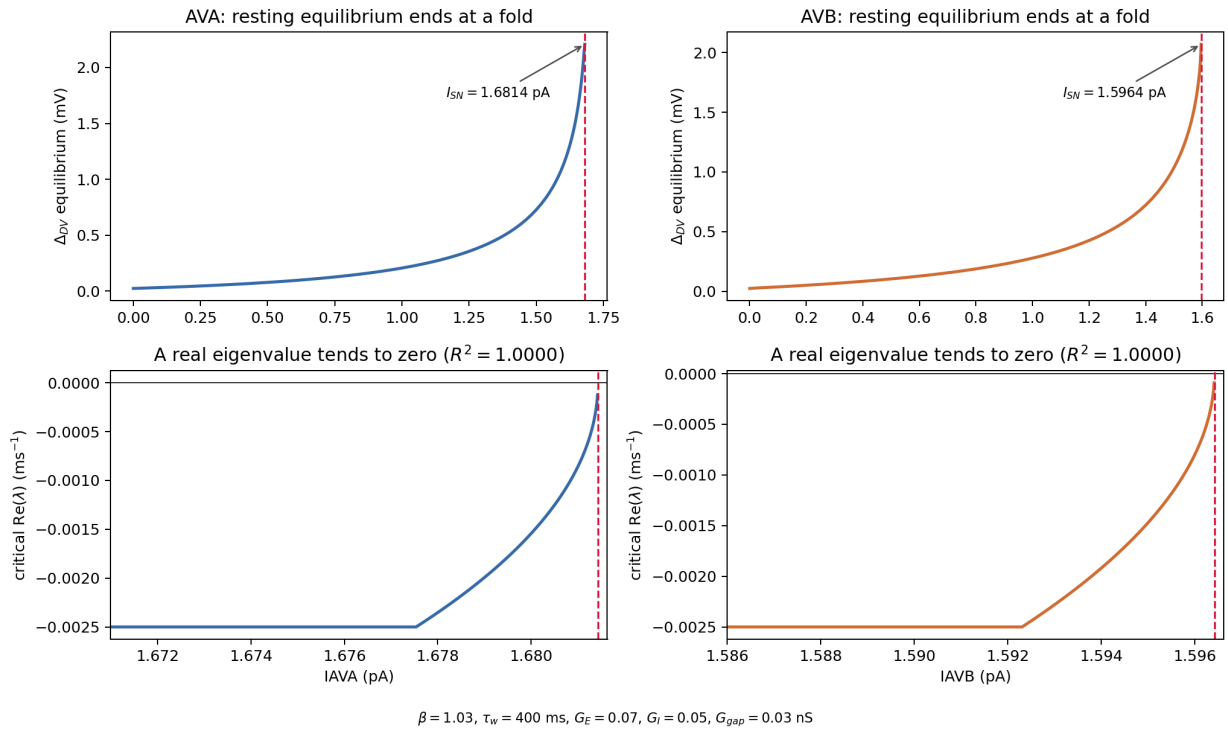


Figure 1: Equilibrium and eigenvalue evidence. The stable resting branch ends at the marked command current. A real eigenvalue, rather than a complex conjugate pair, tends to zero at each fold. The lower panels magnify the last 0.01 pA below onset.

4 Evidence 2: a finite-amplitude cycle has infinite period

Stable periodic activity exists immediately above each equilibrium fold. Its functional dorsoventral excursion remains approximately 48 mV as $\Delta I = I - I_{\text{SN}}$ approaches zero, so the orbit does not collapse onto a small-amplitude oscillation. Instead, the period grows from roughly 3.4 s to 18–19 s over the sampled interval.

The orbit therefore retains its global geometry while spending progressively more time near the location of the vanished equilibria. The fitted law

$$T(\Delta I) = \frac{A}{\sqrt{\Delta I}} + B \quad (3)$$

describes the command branches with $R^2 = 0.99950$ (AVA) and 0.98774 (AVB); see Figure 2. The additive constant B is the regular traversal time around the portion of the orbit away from the saddle-node bottleneck.

AVA and AVB command onset: finite dorsoventral output, infinite period

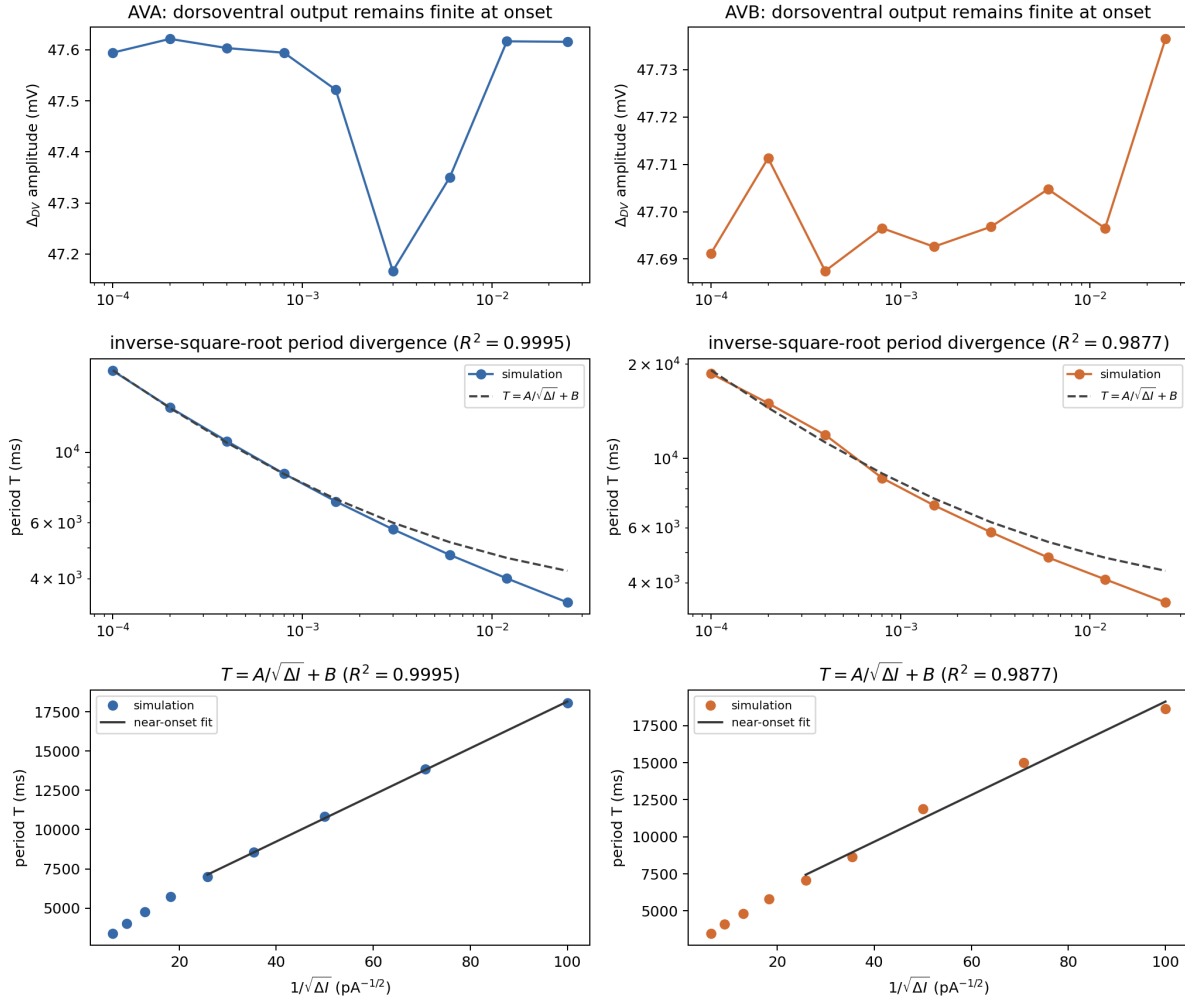


Figure 2: Periodic-orbit evidence. Top: cycle amplitude remains finite near onset. Middle: the period diverges as the drive approaches the fold. Bottom: period is linear in $1/\sqrt{\Delta I}$ over the five closest samples.

5 Why inverse-square-root scaling identifies a SNIC

Near a generic saddle-node, the flow along its center direction has the normal form

$$\dot{x} = \mu + ax^2, \quad \mu = I - I_{\text{SN}}. \quad (4)$$

For $\mu > 0$, the equilibria have disappeared but trajectories still pass slowly through their former location. The bottleneck traversal time is

$$T_{\text{bottleneck}} \sim \int_{-\infty}^{\infty} \frac{dx}{\mu + ax^2} = \frac{\pi}{\sqrt{a\mu}}. \quad (5)$$

Adding the finite time required to traverse the rest of the orbit gives $T = A/\sqrt{\mu} + B$. Thus

$$T \rightarrow \infty, \quad f = \frac{1}{T} \sim \sqrt{I - I_{\text{SN}}} \rightarrow 0, \quad (6)$$

even though the voltage amplitude remains large.

This is precisely what the simulations show. The periodic trajectory passes through the saddle-node bottleneck and closes globally, so the saddle-node lies on the invariant circle traced by the limiting orbit.

6 Why the main alternatives are rejected

Hopf bifurcation. A Hopf transition requires a complex conjugate eigenvalue pair to cross the imaginary axis. Here the critical eigenvalue is real. A Hopf also has a finite linear onset frequency, whereas the measured frequency tends to zero.

Saddle-homoclinic bifurcation. A homoclinic orbit also has an infinite period, but its leading divergence is logarithmic, $T \sim -C \log(\Delta I)$, rather than inverse-square-root. It also does not require the periodic orbit to terminate at the observed equilibrium fold. The measured $T = A/\sqrt{\Delta I} + B$ scaling and fold coincidence support a SNIC instead.

The classification is therefore based on four mutually reinforcing facts: a saddle-node in the equilibrium branch, a real critical eigenvalue, a finite-amplitude cycle immediately above the same current, and the SNIC infinite-period scaling law.

7 Reproduction and outputs

From the repository root, run

```
conda activate simple-worm-scripts
python experiments/drive_snic.py
```

The script generates the two figures used here and `media/bifurcation_drive_snic_results.json`. The JSON record contains every sampled drive, period, frequency, DV amplitude, period method, period CV, fitted fold, and goodness-of-fit statistic.